

Interactive GIS atlas for tidal stream visualization in Hong Kong sailing race planning

Noah Yiu

An interactive GIS project that visualizes predicted tidal stream direction and speed over time to support sailing race preparation in Hong Kong.

[\[View the Interactive Map\]](#)

[\[View the GitHub Repository\]](#)

Overview

I created this project to make tidal information more accessible and useful for sailing race planning in Hong Kong. As an experienced sailor and a former member of the Royal Hong Kong Yacht Club racing team, I wanted to combine sailing with the digital tools we use every day to address a problem I have seen firsthand: before races, there is often limited time to complete pre-race checks, interpret conditions, and come up with an effective race plan.

This project is essentially a more interactive and digitized version of a traditional tidal stream atlas. Instead of relying only on static tables or printed charts, sailors, coaches and race managers can use a map-based interface to explore how tidal stream direction and speed shift across time and location in the race area, making it a more accessible and handy tool for race use.

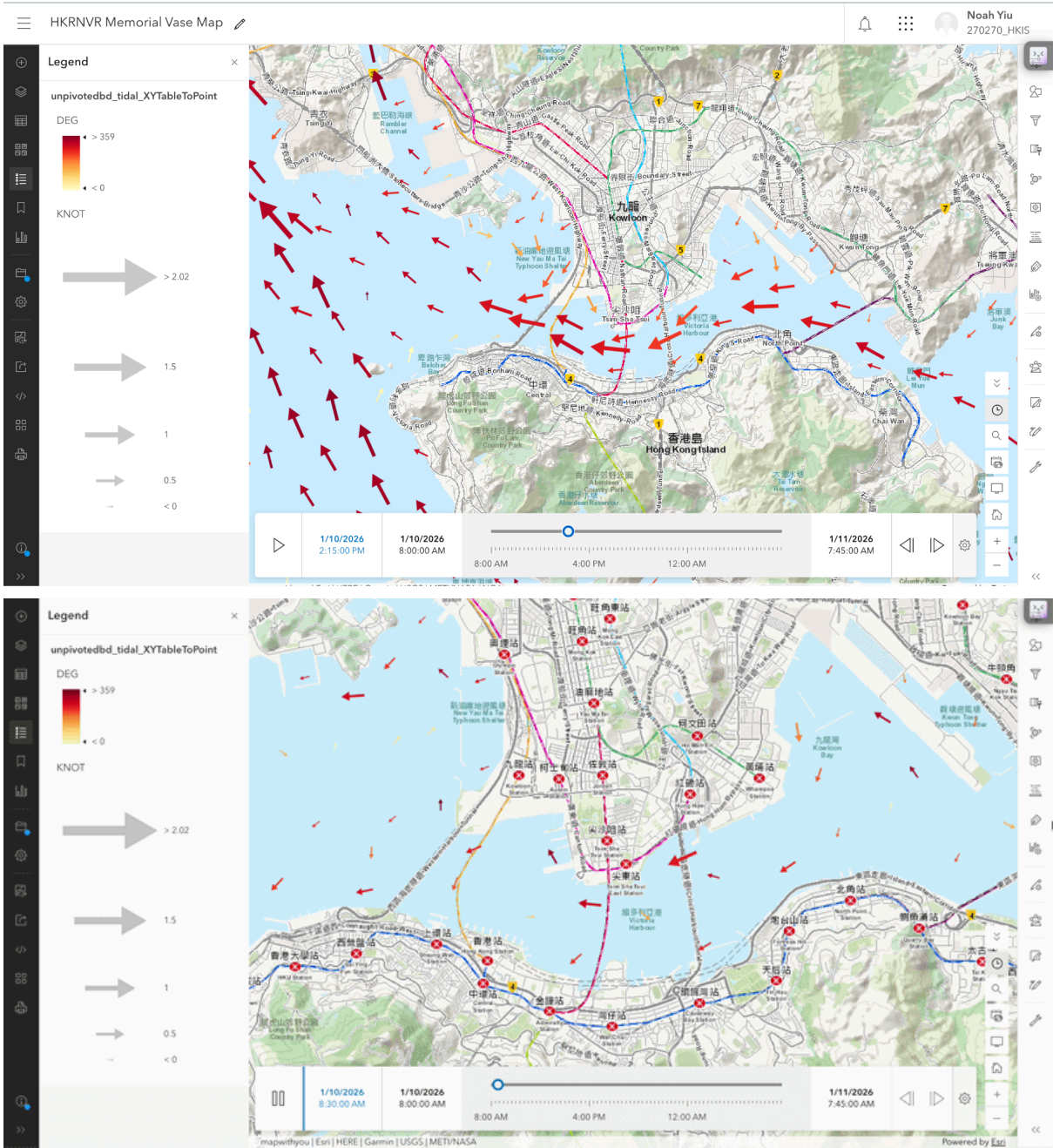
Interactive Map

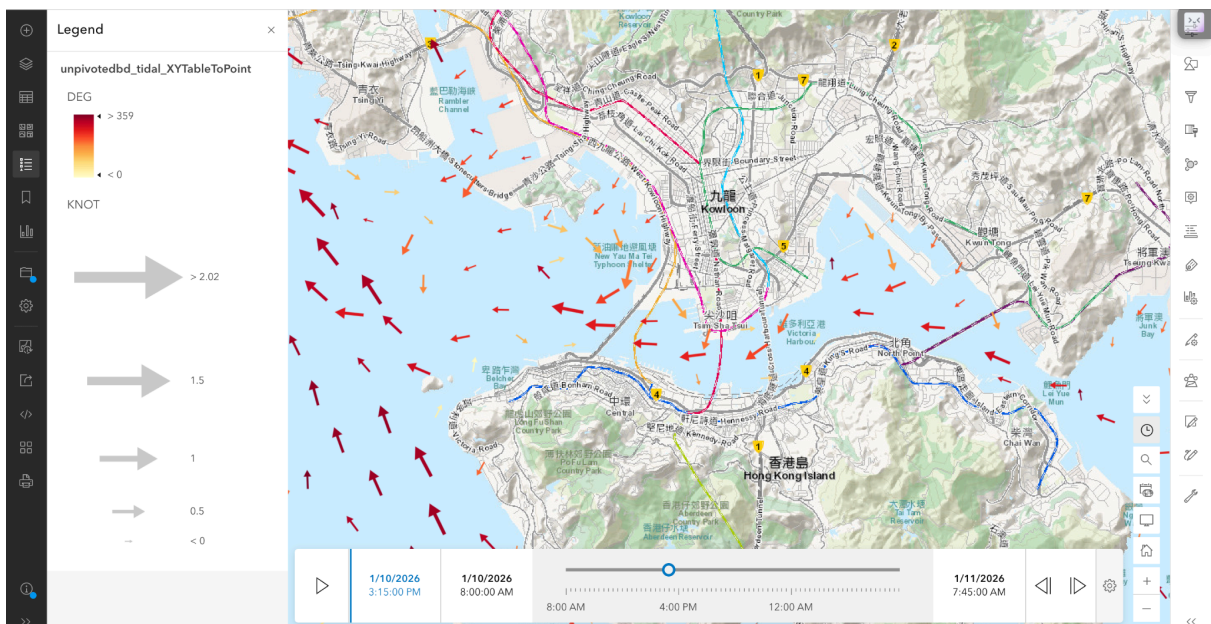
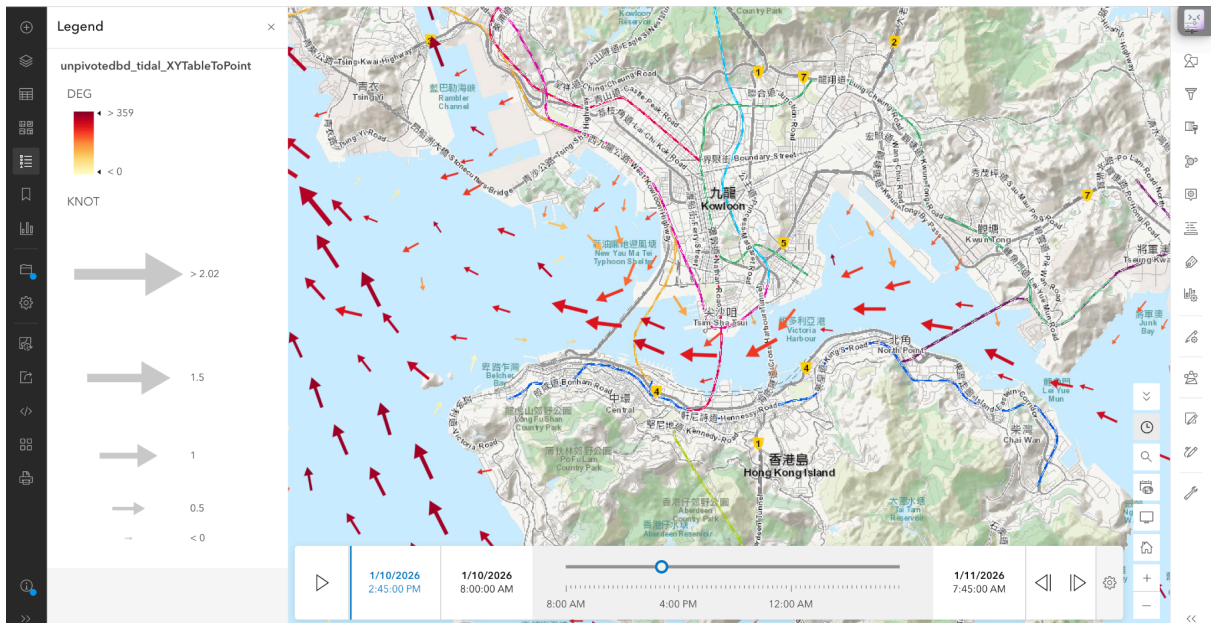
The map is published online through ArcGIS Online and built around a time slider that allows users to animate predicted tidal stream conditions through the race window. In the current prototype, each tidal vector changes with time, and users can click individual vectors to inspect detailed values including latitude, longitude, direction in degrees, and current speed in knots.

The legend explains the visual encoding of the map, including the color ramp and arrow sizing used to represent the tidal stream field. This makes it possible to quickly determine stronger currents, directional shifts, and potentially important local differences near key parts of the race course.

Suggested embed note:

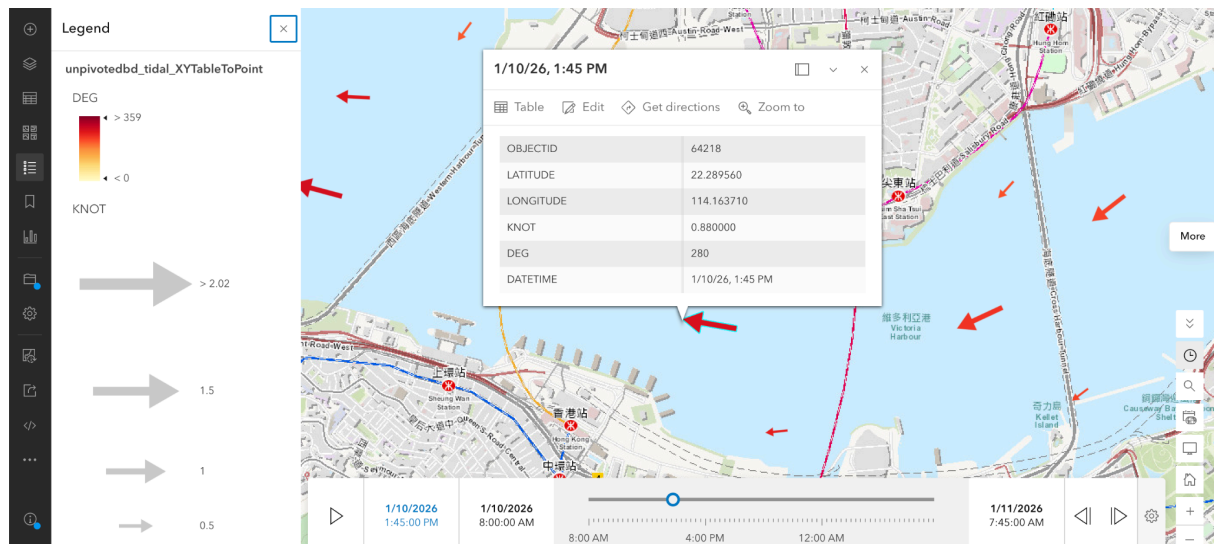
Use the embedded map below, or open it in a new tab for the full ArcGIS Online experience.





Features

As seen in the GIF and Screenshots from above, the darker the vector, the greater the degree and the bigger it is the higher the knot it represents. The time slider shows 15-minute intervals in the changes in each tidal vector. Click on each tidal vector to see it's specific stats, as shown below:



Motivation

This project was developed for a real sailing use case rather than as a purely academic exercise. In racing, small differences in current can affect route choice, mark approaches, starting strategy, and the timing of maneuvers, yet that information is not always easy to interpret quickly in a practical race setting.

My goal was to create a tool that helps make tidal information more intuitive and more immediately usable. By turning raw published tidal prediction data into an interactive spatial interface using GIS, I hoped to build something that could support sailors, coaches, and race organizers in a more practical way.

Method

I downloaded published tidal prediction data from the Hong Kong Tidal Stream Prediction System and imported it into ArcGIS Pro, where I converted the dataset into mapped vector features and configured the map to display both current direction and current speed visually. The project was then published to ArcGIS Online as an interactive web map.

The prototype includes:

- a time-enabled interface for stepping through changing tidal conditions, directional vectors that shift across the forecast window,
- pop-up information for each vector, including coordinate and flow values,
- and a legend for interpreting current strength and direction.

Why it matters

This project sits at the intersection of GIS, marine data visualization, and decision-making in sport. It reflects my interest in using data not just to describe the world, but to build models that make real contributions, helping people make better decisions under time pressure.

More broadly, it shows how geospatial technology can make technical environmental data more understandable in a real operational setting.

Limitations

This prototype is based on published tidal prediction data and should be understood as a tide-based visualization and planning tool. It does not yet incorporate short-term meteorological influences such as wind-driven current changes or a weather-adjusted 96-hour current layer.

Because of that, the current version is best used as a planning and interpretation tool rather than a complete real-time forecasting system.

Next Steps

Possible next steps for the project include:

- incorporating short-term weather-influenced current information where available, [hko](#)
- refining the map for specific race formats or venues,
- adding more race-oriented annotations or decision-support views,
- and improving the interface for faster use before races.

Credits

This project was developed by Noah Yiu.

Special thanks to:

- Ian Fleming, Rear Commodore of the Royal Hong Kong Yacht Club, for project recognition and support.
- Prof. LIU Yan, Professor, Department of Geography and Resource Management, The Chinese University of Hong Kong, for academic guidance in GIS and spatial thinking.
- Adrian Li, Team Coach

Links

- Interactive ArcGIS Online map
- GitHub repository
- Project screenshots